Anastasios A. Manganaris

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EDUCATION

Purdue University, West Lafayette, Indiana — Expected May 2028
Ph.D. Computer Science — GPA: 3.78 (4.0 scale)
Advised by Ahmed H. Qureshi

The Ohio State University, Columbus, Ohio — May 2023
B.S. Computer Science & Engineering, specialization: A. I. — GPA: 3.904 (4.0 scale)
Advised by Sanggu Kim

EXPERIENCE

Research Assistant at Purdue University — May 2024 – Present
Scaling imitation learning, reinforcement learning, and multi-agent task and motion planning to longer horizons and larger robot teams by exploiting problem structure, using formal specifications and hierarchical decompositions.

- Developed Automaton Constrained Q-Learning for temporal-logic-constrained reinforcement learning (NeurIPS 2025)
- Built Graph-of-Constraints MPC for reactive multi-robot task and motion planning, deployed on UR5e arms (ICRA 2026)
- Scaling diffusion policy guidance to compositional tasks using logical world models (submitted to ICRA 2027)
- Developing reactive multi-robot TAMP approaches for loco-manipulator teams, deployed on the Unitree B1 and G1 (ongoing work)

Teaching Assistant at Purdue University — August 2023 – May 2024
Instruction, course organization, development, and grading for Data Structures and Algorithms (CS251, Head TA for CS253)

Research Assistant at The Ohio State University — December 2021 – May 2023
Investigated identifying RNA chemical modifications from Nanopore sequencing data and relationship between Nanopore sequence quality and RNA structures.

Software Development Intern at SentriLock, LLC — June 2018 – August 2018
Designed, implemented, and deployed an application for an analytics dashboard.

PUBLICATIONS

Conference & Journal Papers

A. Manganaris, J. Lu, A.H. Qureshi, S. Jagannathan. Graph-of-Constraints Model Predictive Control for Reactive Multi-agent Task and Motion Planning, IEEE RAS International Conference on Robotics and Automation (ICRA), 2026.

A. Manganaris, V. Giammarino, A.H. Qureshi, S. Jagannathan. Formal Methods in Robot Policy Learning and Verification: A Survey on Current Techniques and Future Directions, Transactions on Machine Learning Research (TMLR), 2025.

A. Manganaris, V. Giammarino, A.H. Qureshi. Automaton Constrained Q-Learning, Conference on Neural Information Processing Systems (NeurIPS), 2025.

Under Review

M. Zoellner, **A. Manganaris**, A.H. Qureshi, R. Paleja. Composing Learned Robot Behaviors with Temporal Logic at Runtime. Under review at IEEE RAS International Conference on Robotics and Automation (ICRA), 2027.

V. Giammarino, **A. Manganaris**, A.H. Qureshi. When Interaction Changes Controllability: Physics-informed Goal-Conditioned Reinforcement Learning for Manipulation. Under review at International Conference on Learning Representations (ICLR), 2027.

Workshop Papers
A. Manganaris, R. Banerjee, G. Prabhakaran, A.H. Qureshi. Riemannian Splat Regression Models for Learning Time Fields on Arbitrary Riemannian Manifolds, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop on Geometric-Aware Representations in Robotics, 2026. **Spotlight Paper**

SERVICE

Co-organizer, IROS 2026 Workshop on Hybrid Architectures for Embodied Autonomy: Bridging Learning, Planning, and Control
Reviewer, IEEE Robotics and Automation Letters (RA-L), 2026 – Present
Reviewer, Conference on Robot Learning (CoRL), 2024, 2026

SKILLS

Robotics: On-policy and off-policy reinforcement learning, diffusion policies, world models, manipulator control, task and motion planning. Experience with the Universal Robots UR5e, Unitree B1, and Unitree G1.
Programming Languages: Python, C/C++, Rust, MATLAB
Software & Tools: ROS, ROS2, Isaac Sim, Isaac Lab, MuJoCo, PyTorch, JAX, Linux, Docker, Git, Slurm, AWS